

Mathieu Castonguay

Gameplay Programmer

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SUMMARY

Gameplay Programmer specializing in movement systems, combat mechanics, and scalable gameplay frameworks in Unity and Unreal Engine. Strong foundation in C++, C#, Blueprints, and GAS, with a focus on clean architecture, player feel, and building responsive, designer-friendly systems.

SKILLS

Programming Languages: C++, C#, Blueprints (UE5), Python, JavaScript, HTML, XML, SQL

Engine & Tools: GIT, Perforce, Unity, Unreal Engine 5, Maya, Visual Studio, Excel, Gameplay Ability System (GAS)

Languages: English (Fluent), French (Fluent)

GAME PROJECTS

Salt & Silver

2025–2026

Gameplay Programmer - (12 persons team – Unreal Engine)

- Implemented a modular combat framework supporting 6 abilities, combo chains, and weapon archetypes using Unreal C++ and Blueprints.
- Architected modular data tables for weapons, abilities, characters, and items, decoupling ability logic from data and accelerating new ability/content creation by roughly 10% across the team.
- Implemented animation blend trees to drive responsive player movement and combat transitions.

Time Lab

2024–2025

Developer & Designer - (Solo – Unity)

- Designed and engineered a first-person puzzle game featuring three synergistic time-manipulation abilities (Rewind, Freeze, Swap).
- Optimized the rewind system's position tracking to update only while objects were in motion rather than every frame, reducing overhead while maintaining 8-step rewind history.

Stunt & Hunt

2024–2025

Game Developer (9 persons team – Unity)

- Created a modular motorcycle physics controller with tunable handling parameters used across multiple levels and enemy encounters.
- Combined physics-based collision with arcade-style feels by reducing in-air gravity and driving trick animations independently of the physics simulation, avoiding the need to hand-tune physics for trick feel.

Plaque Rush

2024–2025

Game Developer (7 persons team – Unity)

- Built mobile input and infinite-runner logic supporting procedural obstacle generation.
- Developed a modular mini-game framework enabling designers to integrate new mini-games

Labyrinth's Legacy

2023–2024

Developer & Designer - (Solo – Unity)

- Implemented procedural terrain generation using noise-based algorithms and runtime mesh modification
- Built a robust save/load system supporting inventory, crafting, and world state persistence.

EDUCATION

University of Central Florida

August 2022 – May 2026

Bachelor of Arts – Digital Media: Game Design

College of Sciences, Nicholson School of Communication and Media